Fiscal Year:	FY 2006	**Task Last Updated:**	FY 07/05/2007
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Project Title:	Development of the otolith system in altered gravity environments		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology (3) Neurobiology		
Space Biology Special Category:	None		
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Comments:	NOTE PI moved to Baylor in Fall 2011; previously at Washington University, St. Louis, MO (Ed., 12/20/2011)		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
Start Date:	02/01/2004	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNA04CC52G		
Performance Goal No.:			
Performance Goal Text:			

In all vertebrates, the otolith system functions to detect head position and movement relative to gravity and to modulate or control gravity depending neuromotor responses. How does gravity influence the development of the otolith receptor cells, their polairzation topography, and their innervating afferents during embryogenesis? The avian otolith system offers an excellent model with which to study the effects of gravity upon development, due to the short maturation period following fertilization and due to the broad existing knowledge of otolith system structure and function in birds. The present project is examining the effects of gravity upon the structure of the avian otolith receptor system. Japanese quails, due to their short incubation time of 15 days are being exposed to different levels of gravity during embryogenesis and their vestibular receptor systems examined using a number of different quantitative anatomical measures. Quail raised from fertilization in either normal gravity, hypergravity (2 - 3G), or microgravity (spaceflight)

Task Description:

conditions will be compared. The anatomical development of the receptor cell organization and the afferent innervation patterns throughout embryogenesis and early post-hatch development are being collected for comparison across the different gravity conditions and developmental stages. The primary goal is to define whether gravity is a determinant trophic factor in development of the otolith system during embryogenesis. In order for manned long duration missions to succeed, the effects of altered gravity environments upon nervous system structure and function must be determined.

The primary objective of the present proposal is to determine the effects of gravity upon development of the peripheral vestibular system. It is known that even moderate exposure to spaceflight conditions in adult animals can produce modifications in the anatomy, physiology, and neuromotor responses related to the vestibular system. In addition, more limited evidence shows that altered gravity environments can directly affect the development of vestibular receptors. Within the vestibular system, the otolith organs comprise the inertial sensors that respond to linear acceleration or tilts of the head relative to gravity. During spaceflight or centrifugation, the acceleration constant due to normal Earth gravity is eliminated or changed, thereby altering the inertial reference upon which vestibular system responses are based. Does gravity serve as a regulatory factor in the development of the vestibular system and its proper neuromotor function? This project serves as a ground-based investigation that will examine vestibular receptor organ development in quails completely reared in either Earth normal or altered hypergravity environments. The current application will build upon the isolated findings of our recent spaceflight investigation on STS-108, where embryogenesis in microgravity was found to affect vestibular otoconia, afferent innervation, and synapse formation in quails raised to embryonic day 12. Current experiments are designed to specifically examine how otoconial biosynthesis, receptor morphology, afferent innervation, and synaptogenesis are affected by exposure to constant hypergravity conditions during embryogenesis and early post-hatch development in Japanese quails.

Objective 1. Determine how altered gravity affects the morphology and molecular biosynthesis of otoconia in birds. Otoconia are dense biomineral particles in the otolith organs that serve as mechanical inertial transducers for the detection of linear acceleration and subsequent receptor cell activation. It is known that in invertebrates and fish, development of the homologous statolith is highly sensitive to altered gravity conditions. We hypothesize that the regulation of otoconial biosynthesis is affected by gravity. In Objective 1, the otoconial morphology and the molecular parameters of otoconial biosynthesis during quail embryogenesis and early post-hatch development will be examined using both normal gravity and hypergravity conditions.

Objective 2. Determine how gravity affects the development of otolith receptor cells, their afferent innervation, and synaptogenesis. Two types of receptor cells, three afferent types, and two major forms of hair cell-afferent synapses are known to exist in the otolith sensory epithelium of all amniote vertebrates. These morphological characteristics develop at different times during embryogenesis and post-hatch maturation. We hypothesize that the development of receptor cell distribution, afferent innervation, and synaptogenesis in the otolith system is regulated by gravity. We predict that the rate of maturation, number of cells, number and type of synapses, and afferent terminal innervations will all increase proportionally with the force of gravity. In Objective 2 we will examine these morphological characteristics at different developmental ages in quails raised in either normal or hypergravity environments.

Rationale for HRP Directed Research:

In all vertebrate animals, the vestibular system forms an essential component in the production of movement related responses that are critical for the daily function and survival of the animal. During manned spaceflight, vestibular disturbances frequently occur, with approximately 80 – 90 % of current flight crew personnel experiencing some symptoms space motion sickness, such as disorientation, nausea, and emetic attacks during the first 48 - 72 hours of weightlessness (Davis et al., 1988; Homick et al., 1984; Reschke et al., 1998; Thorton et al., 1987). These symptoms can best be described as acute, although they often return after the flight immediately upon reintroduction to the 1g environment. However, more serious vestibular related effects that occur during long duration spaceflight include changes in compensatory eye movements (Clarke et al., 1993; Clement and Berthoz, 1990; Cohen et al., 1992; Kornilova et al., 1993; Yakovleva et al., 1982), muscle tone and reflex (Edgerton and Roy, 1995; Edgerton et al., 1995), orthostatic intolerance (Yates and Kerman, 1998), and postural stability (Anderson et al., 1986; McDonald et al., 1994; Paloski et al., 1992). Although a number of investigators have postulated that the lack of gravity as a constant stimulus during spaceflight produces profound changes in vestibular system function, the basic mechanisms that are affected by changes in gravity are only beginning to be addressed. Several reports have indicated that the structure and function of the vestibular system is affected by exposure to altered gravity conditions. For example, changes in size of otoconia (Fermin et al., 1996; Ross and Donovan, 1986; Wiederhold et al., 2000), in the receptor-afferent morphology (Ross, 1993; 2000; Gaboyard et al., 2002; Raymond et al., 2000), in hair cell conductance (Chabbert et al., 2003), in vestibular afferent responsiveness (Boyle et al., 2001; Bracchi et al., 1975; Correia et al., 1992; Jones et al., 1993), in vestibular central pathways (Anken et al., 2002; Bruce and Fritzsch, 1997; Kozlovskaya et al., 1984; Pomeiano et al., 2002), and in vestibular related neuromotor responses (Dai et al., 1994; Sebastian, 1996; Takabayashi et al., 1987; von Baumgarten et al., 1975) have all been observed in both adult and developing animals exposed for brief periods in either microgravity or hypergravity. The current project seeks to determine how gravity affects the morphology of the vestibular otolith receptor system during embryogenesis and post-hatchling development. Before long duration missions such as a manned presence on the Moon or a Mars flight can be manifested in Spirals 4 and 5, basic knowledge regarding the nature and extent of developmental or adaptational changes in sensorimotor systems that occur as a consequence of exposure to altered gravity conditions must be acquired.

Research Impact/Earth Benefits:

Many sensory systems are known to enter a critical period of development, during which stimulation of modality specific receptors is required for normal neural function. For example, in the visual system, if deprived of light during the early critical period, normal sight is impaired or eliminated during maturation. In the vestibular system which controls normal postural, eye and head movement responses during motion, no effort to study the effects of gravity upon development have been extensively performed. Thus, it is currently unknown whether altered gravity environments affect vestibular system development and subsequent reflex motor behavior. The current project is examining the trophic influence of gravity upon the development of receptors and their innervating afferent neurons in qauil embryos. Both hypergravity as induced by chronic centrifugation and microgravity as induced during spaceflight are being used to determine the morphological development of vestibular organs at different stages of development. The research should provide basic answers into questions related to vestibular receptor pathology that is known to occur with aging humans and in some disease cases.

Task Progress:

Centrifuge construction: In order to study the effects of altered gravity environments upon development of vestibular structure and function, we needed to design and build a habitat for hypergravity exposure. We have developed an 8 foot radius centrifuge capable of housing as many as 6 pregnant female mice at hypergravity and 4 mice in control rotational non-hypergravity conditions. The habitat features an enclosed chamber with automated feed, water, lighting, and temperature regulation system. There are 5 video cameras to record behavior, and a computer interface to monitor environmental conditions. A four chamber control habitat is mounted on the central rotation axis to deliver identical rotational exposure without hypergravity (coriolis force control). The centrifuge is housed inside a completely closed circular chamber that is temperature regulated and safety-locked for access. The design and building of the hypergravity exposure unit was involved and detailed.
Consultation with NASA engineers provided information that reveals that our facility is only one of four in the United States capable of providing chronic hypergravity exposures from fertilization to sexual maturity in small animal species. The other three are located in NASA facilities. These are landmark studies and should provide experimental environments capable of answering fundamental questions regarding critical period formation for vestibular system development and beyond.

Morphology and biosynthesis of otoconia: Our first project objective was to investigate the biosynthesis of otoconia in quails developed in normal and hypergravity environments. Using an in-situ probe developed in our laboratory, we found that the major otoconial protein, termed otoconin 90 (OC90) is expressed in quail. We examined Oc90 mRNA expression from stage E2 through post-hatch day P7 and in adults. Oc90 was expressed in the thin, nondifferentiated epithelia on the roof of saccule, utricle, and lagena but was absent from the sensory neuroepithelia. Interestingly, this unique epithelial expression pattern of Oc90 mRNA in the quail is virtually identical to that of mammals. We observed increasing expression of Oc90 at E2 – E8, with a peak expression of E9, which declined from E9 – E12. By E13 – P7, no or very little expression was observed. These findings suggest that otoconial formation occurs as an interaction of secreted OC90 with other local factors in the gelatinous membrane (Wang et al., 1998; Thalmann et al., 2001). In support of that hypothesis, a novel gene, Otopetrin (Otop1) has been identified as the point mutation source for the naturally occurring mutant tilted (tlt) and mergulhador (mlh) mice that lack otoconia (Hurle et al., 2003). We have also examined Otop1 mRNA expression in developing quail vestibular organs using an in-situ hybridization probe developed in our lab. Otop1 was expressed in the otolith receptor epithelium, most likely in the support cells. Next, we developed quail embryos in a constant hypergravity 2g environment from fertilization. Quantitative rtPCR was performed using cDNAs for both Oc90 and Otop1 for each developmental day throughout embryogenesis (E2 – E15) and at several post-hatch time points. Control expression of GapDH was also performed. We then compared the expression levels of these important otoconial production genes for animals raised in both 1g and 2g conditions, in order to determine if gravity affects the expression of genes that regulate otoconial growth. Oc90 showed a similar expression profile to Otop1 during development; production of Oc90 reached a maximum at E9 and decreased toward hatching. However, the rate of this regression was lower than Otop1, suggesting that Oc90 may be an active player during growth and maintenance of mature otoconia. Oc90 level was significantly lower in brain than inner ear. Expression of Otop1 was increased at E7 and E10 by hypergravity, but for other time points it did not change significantly. Similar lack of effect due to hypgergravity exposure was observed for Oc90. These quantitative results showed that the effect of gravity as an epigenetic determinant of maturation and growth of otoconia is relatively weak. Accordingly, we also measured otoconial calcium weight throughout development and have not observed any significant difference between animals grown at normal and hypergravity conditions. We are currently counting the number, size and distribution of otoconial particles as well as measuring the mass of the entire stone sheet to determine if the observed lack of gene expression is consistent with morphological measures.

Morphology of avian otolith receptors, afferent innervation, and synaptogenesis: In Objective 2, we hypothesize that the development of vestibular receptor cell distribution, afferent innervation, and synaptogenesis in the otolith organs are dependent upon gravity. Our first project objective was to investigate the morphology of otoconia in quails developed from fertilization in normal and hypergravity environments. Japanese quails have a 15 day incubation time. The otoconia of Japanese quails developed in Earth 1g environments were first examined at each embryonic stage from E4 – E15 using scanning electron microscopy (SEM) and sectioned tissue histology. At developmental stage E4, otoconia were not yet detectable, but at stage E5 immature otoconia were beginning to form in the saccule but not the utricle or lagena. At E6, otoconia began to form in the utricle and lagena. Most of the newly developing otoconia in quails were small with double fluted ends. A quantitative analysis of the number and size of otoconia for animals raised in 1g conditions was performed, with measures taken at each embryonic day and at each of the first four post-hatch weeks. Otoconia from different regions of the maculae were compared, as it is known that otoconia vary in size depending upon macular location in adults. We implemented two quantification methods for examining otoconial growth. The first, provided measurements of the length and width of individual otoconia. Values for over were obtained from each of 3 standard locations (i.e., lateral striola, central, medial periphery) in each otolith organ, counting only otoconia that lay approximately perpendicular to the visualization plane. Over 13,000 otoconia were quantified for the utricle and saccule, at the different developmental stages. For example, at E10 a total of 976 otoconia from 4 utricles were measured; where sizes ranged between 0.9 – 19.7 micrometers in length, 0.8 – 8.2 micrometers in width, and mean length values (±SD) were 8.0 (±3.6) for striolar, 6.6 (±2.8) for central, and 6.6 (±2.6) for medial peripheral regions. In adult quails, the otoconia were significantly larger. Comparable values for 596 otoconia from adult (>6 week) quails were: length ranged between 1.6 – 45.5 µm, width ranged between 1.2 – 19.7 µm, and mean length values of 18.0 (±9.6) µm for striola, 10.2 (±4.8) µm for central, and 9.1 (±3.3) µm for medial peripheral regions. Mature individual otoconium were characterized by a barrel shape with two trihedral faceted ends. However, initial formation of otoconia at E5 (saccular) and E6 (utricular) maculae was characterized by a double fluted morphology that consisted of an hourglass shape with extended fins forming trihedral angles of 120º. Double fluted otoconia rapidly filled, so that by embryonic day 8 mature otoconia dominated the maculae for the remainder of development through adulthood. Thus, a progression from double fluted to mature forms was noted. Mature utricular otoconia in adult quails averaged 11µm in length and 5µm in width, with length/width ratios of approximately 2.5:1, for all size ranges. Saccular otoconia were smaller, having about 70% the size of utricular otoconia in both length and width. During development, the average size and range of individual otoconia increased nearly linearly for both otolith organs. In the utricular macula, large otoconia were concentrated in the lateral regions of the epithelium. In contrast, otoconia of various sizes were distributed uniformly across the surface of the saccular macula.

We have also initiated studies to examine the biosynthesis of otoconia during development. Using an in-situ probe developed in our laboratory, we found that the major protein contained in mammalian and avian otoconia, termed otoconin 90 (OC90) is expressed in quail. We examined Oc90 mRNA expression from stage E4 through post-hatch day P7. Oc90 was expressed in the thin, nondifferentiated epithelia on the roof of saccule, utricle, and lagena but was absent

from the sensory neuroepithelia. Interestingly, this unique epithelial expression pattern of Oc90 mRNA in the quail is virtually identical to that of mammals. We observed strong expression of Oc90 at E4 – E8, with declining expression at E9 – E12. By E13 – P7, no expression was observed. These findings suggest that otoconial formation occurs as an interaction of secreted OC90 with other local factors in the gelatinous membrane (Wang et al., 1998; Thalmann et al., 2001). In support of that hypothesis, a novel gene, Otopetrin (Otop1) has been identified as the point mutation source for the naturally occurring mutant tilted (tlt) and mergulhador (mlh) mice that lack otoconia (Hurle et al., 2003). In wild type mice, Otop1 mRNA was found using in-situ hybridization to be expressed in the otolith receptor epithelium, most likely in the support cells, and OTOP1 was identified as a protein found in the acellular matrix of the otoconial membrane in the utricle and saccule. Recently, three separate highly conserved regions of the murine Otop1 cDNA have been identified in expressed sequence tag (EST) clones of chicks, strongly suggesting similar mRNA and protein are also expressed in quails.

Finally, we conducted an initial study to document the morphological development of the avian otolith vestibular receptor system during embryonic and post-hatch maturation. We focused on four aspects of development which were crucial to otolith system function including, epithelial growth, hair cell density, hair cell stereocilia polarization, and afferent nerve innervation. We found that despite occasional plateaus, the growth curves for both the utricle and saccule increased in a sigmoidal fashion throughout embryonic development before leveling out after P7. Hair cell density varied as the embryo developed and was dependent upon macular location. The complete pattern of hair cell morphological polarization across both maculae in the adult was observed using scanning electron microscopy. This pattern was first established at E8, the halfway point of embryonic development. Although other bundle morphologies did exist, the vast majority of immature hair cells first appeared in the sensory epithelia fully polarized with the kinocilia located at the periphery of the stereocilia bundle. The number of immature hair cells decreased as the embryo developed but new cells were still apparent in the mature otolith maculae. Using immunohistochemistry and tract tracing techniques, the shape and location of the striolar zones in both otolith endorgans was delineated. Prior to E10, only type II hair cells populate the maculae and only bouton afferents innervate the epithelium. At E10, the first few digitiform protocalyces appear that gradually develop into mature calyceal terminals by E12. At E12, dimorph and calyx afferents are beginning to populate the striolar regions. Finally, the topographic organization of afferent macular innervation in the adult quail utricle was examined. Taken together, these results form a comprehensive picture of the otolith receptors of Japanese quail throughout development. These studies are now being performed for mice. We have chosen the JAX labs 128-SJV strain, as they are gentle animals, good breeders, pigmented, and carry no know deficits in hair cell, inner ear, or visual dysfunction.

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